

Renormalization-group approach to antiferromagnetic critical behavior

V. A. Alessandrini, H. J. de Vega, and F. Schaposnik
Departamento de Física, Universidad Nacional de La Plata, La Plata, Argentina
 (Received 19 March 1974)

The behavior of an antiferromagnetic system near the critical point is discussed in the context of Wilson's theory of critical phenomena. Using the two-lattice model to describe the antiferromagnetic phase transition, it is shown that the critical behavior is the same as that of a ferromagnetic system. The approach based on renormalized Lagrangian field theory is also discussed.

I. INTRODUCTION

There has recently been much interest in studying the infrared behavior of Euclidean Lagrangian field theories of the $\lambda\phi^4$ type in $(4-\epsilon)$ -space dimensions.¹⁻⁴ If the scalar field has N components ϕ_i ($i=1, \dots, N$), the infrared behavior of such a theory describes the critical behavior of a classical Heisenberg ferromagnet. These theories, within the context of Wilson's ϵ expansion,⁴ provide us with a nontrivial example of the phenomenon of non-canonical scaling which, needless to say, is most welcome in statistical mechanics, where critical indices in general are different from those predicted by mean-field theories (canonical scaling).

The purpose of this paper is to show that a simple modification of Wilson's theory of critical phenomena⁴ can account for antiferromagnetic critical points as well. The two-lattice model of an antiferromagnetic system⁵ leads in the continuum limit to a Lagrangian field theory consisting of two coupled N -component scalar fields. In order to describe antiferromagnetic critical behavior, one must demand that the mass of one of the fields vanishes at the critical temperature, while the mass of the other one remains finite. The field that has a vanishing mass at the critical temperature is associated with the staggered magnetization, while the other one is associated with the physical magnetization of the system. Not surprisingly, we shall find that the critical behavior of such an antiferromagnetic system is entirely equivalent to the critical behavior of the classical N -component ferromagnet.

In this paper we shall restrict ourselves to the Ising-like case ($N=1$), although it will be obvious that the discussion is valid for N arbitrary. We shall first discuss the trivial Gaussian model to illustrate the simple ideas at work. Then we shall proceed to discuss the effect of the non-Gaussian terms (interactions) that appear when one makes more realistic approximations, both in the context of Wilson's version of the renormalization group⁴ and by using the methods of renormalized Lagrangian field theory.⁶

II. THE GAUSSIAN MODEL

Let us start by considering an Ising-like model consisting of two spin lattices s_i with couplings J_{ij} ($i, j=1, 2$). After making the usual approximations of taking the continuum limit and smoothing the δ functions $\delta(s_i^2 - 1)$ by replacing them with Gaussian probability distributions of the form $\exp(-\frac{1}{2}bs_i^2)$,⁴ the partition function for the system is written as

$$Z = \int \mathcal{D}\phi_1 \mathcal{D}\phi_2 \exp\{\mathcal{H}[\phi_1, \phi_2]\}, \quad (1)$$

with a Hamiltonian given by

$$\mathcal{H}[\phi_1, \phi_2] = -\frac{1}{2} \sum_{i,j} \int d\vec{q} \phi_i(\vec{q})(K_{ij}q^2 + r_{ij})\phi_j(-\vec{q}). \quad (2)$$

We have introduced the parameter $K_{ij} = J_{ij}/kT$. Moreover, the r_{ij} are given by

$$\begin{aligned} r_{11} &= b - 2dK_{11}, \\ r_{22} &= b - 2dK_{22}, \\ r_{12} &= -4dK_{12}, \end{aligned} \quad (3)$$

where d is the dimensionality of the lattice. An ultraviolet cutoff Λ , of the order of the inverse of the lattice spacing, is understood in Eq. (2).

If one now performs a renormalization-group transformation according to Wilson⁴ by first integrating the high-frequency components of the fields ϕ_i in the partition function and then rescales the momenta and fields in such a way that the new Hamiltonian has the same kinetic-energy part as the old one, one gets the usual Gaussian fixed point $r_{ij}=0$ which implies, according to Eq. (3), $K_{11}=K_{22}$ and $K_{12}=0$. We therefore find the conventional model for a two-component ferromagnetic system, with the internal symmetry $O(2)$.

In order to describe antiferromagnetic behavior, one needs symmetry under the interchange $1 \rightarrow 2$ ("charge" symmetry instead of "isospin" symmetry). This implies that $K_{11}=K_{22}=K$, $K_{12}=K'$, and therefore, by Eq. (3) it follows that $r_{11}=r_{22}=m$, $r_{12}=m'$. In this condition the quadratic form

in the Hamiltonian can be diagonalized for any value of q^2 . Introducing the new fields (sums and difference of sublattice magnetizations)

$$\phi_{\pm}(q) = (1/\sqrt{2})[\phi_1(\vec{q}) \pm \phi_2(\vec{q})] \quad (4)$$

the Hamiltonian can be written in the form $\mathcal{H} = \mathcal{H}_+ + \mathcal{H}_-$, where

$$\mathcal{H}_{\pm} = -\frac{1}{2} \int d\mathbf{q} \lambda_{\pm}(q) \phi_{\pm}(q) \phi_{\pm}(-q) \quad (5)$$

and

$$\lambda_{\pm}(q) = (m \pm m') + q^2(K \pm K') \quad (6)$$

Since the two fields $\phi_{\pm}$ decouple, the partition function Z becomes the product of the two partition functions $Z_+ Z_-$. It is now clear that we can perform a renormalization-group transformation only on the field ϕ_+ , obtaining the fixed point

$$m + m' = 0, \quad (7)$$

or on the field ϕ_- , obtaining the fixed point

$$m - m' = 0. \quad (8)$$

Since $m \pm m'$ are the eigenvalues of the mass matrix $\{r_{ij}\}$, both conditions (7) and (8) are equivalent to $\det\{r\} = 0$. If we perform a renormalization-group transformation on both fields $\phi_{\pm}$, then both eigenvalues are zero and we obtain $m = m' = 0$, the two-component ferromagnetic fixed point.

The antiferromagnetic critical point is reached by performing a renormalization-group transformation on only one of the two fields, $\phi_-(x)$. All the results of Néel's two-sublattice mean field theory⁵ are trivially obtained in this way. By making use of Eq. (3) we find that Eqs. (7) and (8) determine two critical temperatures Θ and T_N , respectively. Taking $J > 0$, but $J' < 0$, one easily finds

$$\Theta = (2d/kb)(J - 2|J'|), \quad (9)$$

$$T_N = (2d/kb)(J + 2|J'|). \quad (10)$$

Then we see that Néel's temperature T_N is larger than Θ , and that it is the critical temperature corresponding to the antiferromagnetic phase transition.

The magnetic susceptibility χ is given in terms of the correlation function $G_{ij}(p) = \langle \phi_i(p) \phi_j(-p) \rangle$ by

$$\chi = \sum_{i,j=1}^2 G_{ij}(p=0), \quad (11)$$

so, near T_N , χ takes the form ($A = K^2 - K'^2$)

$$\chi = \frac{2}{Ab(m+m')} \simeq \frac{1}{1 + (\Theta/T)} \quad (12)$$

and is therefore finite, as it should be.

III. THE INTERACTING CASE

A more realistic smoothing of the δ function that ensures that the spin fields $\phi_i(\vec{x})$ take only the values ± 1 leads to a probability distribution of the form of an exponential of quadratic and quartic terms in the fields. One must therefore add to the Hamiltonian of Eq. (5) an interaction Hamiltonian of the form

$$\mathcal{H}_I = u_0(\phi_1^4 + \phi_2^4) = \frac{1}{2} u_0(\phi_+^4 + \phi_-^4 + 6\phi_+^2\phi_-^2), \quad (13)$$

and consequently the two fields ϕ_+ , ϕ_- are coupled, the partition function no longer factorizes as in the Gaussian case, and the concept of the partial renormalization-group transformation is no longer meaningful. From now on we change notation and denote by $\phi(\vec{x}) = \phi_-(\vec{x})$ the field that, in the Gaussian case, has zero mass at the Néel temperature and by $\chi(\vec{x}) = \phi_+(\vec{x})$, the field that still retains a mass at the antiferromagnetic critical point. The field $\phi(\vec{x})$ develops a nonvanishing vacuum expectation value—the staggered magnetization—below the Néel temperature even in the absence of an external magnetic field. The mass squared of the fields ϕ and χ will be denoted by $r = m - m'$ and $R = m + m'$, respectively.

As pointed out before, we must now perform a renormalization-group transformation on both fields. We know that the two-component ferromagnetic case corresponds to the situation in which both masses, r and R , vanish at the critical temperature. We also know from the work of Brezin, Zinn-Justin, and Le Guillou on multicomponent systems⁷ that, even in the case of the most general quartic interactions among the two fields, the large-distance behavior of the correlation functions is controlled by the usual two-component ferromagnetic fixed point with $O(2)$ symmetry, in spite of the fact that the original Hamiltonian does not in general exhibit this internal symmetry.

There are, however, other fixed points of the renormalization-group transformations (RGT). It is instructive to remember what happens in the simplest possible case, the one-component Ising model discussed at length by Wilson and Kogut.⁴ The asymptotic behavior of the correlation functions for $r \ll p^2 \ll \Lambda^2$ —that is to say, distances large compared to the lattice spacing but small compared to the correlation length of the system—is controlled by a fixed point $r^* = 0(\epsilon)$ and $u^* = 0(\epsilon)$ in $4 - \epsilon$ dimensions, u being the coupling constant corresponding to the quartic self-coupling of the field. As long as we are in the vicinity of the critical temperature and the momenta are in the above-mentioned region, the effective coupling constants (the parameters of the transformed Hamiltonians) are in the vicinity of this fixed point. However, this fixed point is not fully attractive but rather a saddle

point. When $p^2 \approx r$ the renormalization-group trajectories begin to depart from it, and as p^2 becomes much smaller than r , for distances larger than the correlation length, they approach another fully stable fixed point corresponding to $r = \infty$. Both kinds of fixed points are in general expected to be associated with any coupling constant with the dimension of mass.⁴

We know from our previous discussion of the Gaussian model that at the antiferromagnetic critical point only one of the masses— r in our case—should have a fixed point of order ϵ in $4 - \epsilon$ dimensions. When the fields are coupled the remaining mass R must necessarily become infinite if it does

not vanish, because in order to have scale invariance all dimensional coupling constants must disappear from the theory. We expect therefore to find a scale invariant regime in the region $r \ll \vec{p}^2 \ll \Lambda^2$, R , and the behavior of the correlation functions in this region will be controlled by a fixed point corresponding to $r^* = O(\epsilon)$ in $4 - \epsilon$ dimensions and $R^* = \infty$.

One can now discuss the antiferromagnetic critical behavior using Wilson's approach based on the use of bare perturbation theory with a cutoff Λ equal to 1; this just amounts to the definition of the unit of mass. The Hamiltonian reads, in d -space dimensions,

$$\mathcal{H} = -\frac{1}{2} \int_{|\vec{q}| < 1} d^d q (q^2 + r) \phi(q) \phi(-q) - \frac{1}{2} \int_{|\vec{q}| < 1} d^d q (q^2 + R) \chi(q) \chi(-q) - \int_{|\vec{q}_i| < 1} \prod_{i=1}^4 d^d q_i \delta^{(d)} \left(\sum_{i=1}^4 \vec{q}_i \right) [u \phi(q_1) \phi(q_2) \phi(q_3) \phi(q_4) + g \phi(q_1) \phi(q_2) \chi(q_3) \chi(q_4) + v \chi(q_1) \chi(q_2) \chi(q_3) \chi(q_4)] , \quad (14)$$

where we have allowed for general values of the quartic coupling constants because fixed points of the RGT do not depend in principle on the initial values of the coupling constants involved in the transformation. However, this statement is true but requires some qualifications. When there are several fixed points the solutions of the renormalization-group equations for the effective masses and coupling constants will depend on the initial Hamiltonian to some extent. Each fixed point has its own domain of attraction in the space of interactions and, given an initial Hamiltonian, the recurrence relations for masses and coupling constants generated by the RGT will have different solutions according to which one is the fixed point in whose range of attraction that interaction is.⁴

How can we make sure that we are inside the domain of the antiferromagnetic critical point $r^* = 0$, $R^* = \infty$? The answer seems obvious—one should take $r \ll \Lambda^2$, $R \sim \Lambda^2$. Since each RGT multiplies R by a factor 4 (this is strictly true only in the Gaussian case) after a few iterations R will be much bigger than the cutoff $\Lambda = 1$ and well within the range of attraction of the fixed point $R^* = \infty$. Then in the free Hamiltonian for the field, the mass term $R \chi(q) \chi(-q)$ is the dominant one and its kinetic energy can be neglected. In order to discuss this fixed point it is convenient to perform the field transformation

$$\psi(q) = R^{1/2} \chi(q) \quad (15)$$

and write the Hamiltonian as

$$\mathcal{H} = -\frac{1}{2} \int_{|\vec{q}| < 1} d^d q (q^2 + r) \phi(q) \phi(-q) - \frac{1}{2} \int_{|\vec{q}| < 1} d^d q (1 + q^2/R) \psi(q) \psi(-q) - \int_{|\vec{q}_i| < 1} \prod_{i=1}^4 d^d q_i \delta^{(d)} \left(\sum_{i=1}^4 \vec{q}_i \right) [u \phi(q_1) \phi(q_2) \phi(q_3) \phi(q_4) + \bar{g} \phi(q_1) \phi(q_2) \psi(q_3) \psi(q_4) + \bar{v} \psi(q_1) \psi(q_2) \psi(q_3) \psi(q_4)] , \quad (16)$$

where

$$\bar{g} = gR^{-1} , \quad \bar{v} = vR^{-2} . \quad (17)$$

Now we can look at the static limit and neglect the kinetic-energy part of the field $\psi(q)$. It will then have a dimensionless propagator equal to 1. The

RGT is performed according to Wilson in the usual way, by integrating the high-frequency components of the fields ϕ , ψ in the path integral representation of the partition function⁴

$$Z = \int \mathcal{D}\phi \mathcal{D}\psi \exp \{ \beta \mathcal{H}[\phi, \psi] \} , \quad (18)$$

and then rescaling all momenta in the low-frequency region $0 < |\vec{q}| < \frac{1}{2}$ to the original interval $0 < |\vec{q}| < 1$; and rescaling also the fields in the following way:

$$\bar{\phi}(p) = \xi \phi(2p), \quad (19)$$

$$\bar{\psi}(p) = \eta \psi(2p). \quad (20)$$

The wave-function renormalization constant ξ is determined by requiring that the kinetic-energy part of the ϕ field be invariant under the RGT.⁴ This requirement ensures that $r=0$ is a fixed point to the lowest order of perturbation theory (Gaussian case) and gives

$$\xi = 2^{1+d/2}. \quad (21)$$

However, since we want $R=\infty$ to be a fixed point for the field ψ , we immediately see by looking at the Hamiltonian as written in Eq. (16) that we must

determine η by requiring that the mass term, the 1 in the free Hamiltonian, be invariant under the RGT. We then obtain, also for the Gaussian case

$$\eta = 2^{d/2}. \quad (22)$$

Finally, the integration over the high-frequency components $\frac{1}{2} < |\vec{q}| < 1$ of the fields is performed by using perturbation theory in which case it reduces to the calculation of Feynman diagrams of an Euclidean field theory in d -space dimensions, with suitably determined weight factors, and with the following qualifications⁴: (i) Internal lines carry momenta only in the high-frequency range and (ii) External lines have momenta in the low-frequency range.

As an example, the one-loop contribution due to internal lines of the field ϕ to the effective quadratic coupling of four ϕ fields is of the form

$$-36u^2 \int_{0 < |\vec{q}_i| < 1/2} \prod_{i=1}^4 d^d q_i \delta^{(d)}\left(\sum_{i=1}^4 q_i\right) \phi(q_1) \phi(q_2) \phi(q_3) \phi(q_4) \int \frac{d^d p}{(p^2 + r)[(q_1 + q_2 - p)^2 + r]}, \quad (23)$$

while its contribution to the effective $\phi^2\psi^2$ coupling is

$$-6u\bar{g} \int_{0 < |\vec{q}_i| < 1/2} \prod_{i=1}^4 d^d q_i \delta^{(d)}\left(\sum_{i=1}^4 q_i\right) \phi(q_1) \phi(q_2) \psi(q_3) \psi(q_4) \int \frac{d^d p}{(p^2 + r)[(q_1 + q_2 - p)^2 + r]}. \quad (24)$$

After rescaling momenta and fields, Eqs. (23) and (24) develop in front factors of 2^{4-d} and 2^{2-d} , respectively, as a consequence of Jacobians and wave-function renormalization constants. The difference is entirely due to the different values of ξ and η , as given by Eqs. (21 and (22), because, up to the order of perturbation theory we are considering here, we only need their Gaussian values.

Approximate recurrence relations for the parameters of the theory can easily be obtained by making the same approximations as Wilson and Kogut on the Feynman integrals.⁴ The recurrence relation reads

$$r_{i+1} = 4 \left(r_i + 3c \frac{u_i}{1+r_i} + \frac{1}{2} c \bar{g}_i \right), \quad (25)$$

$$u_{i+1} = 2^{4-d} \left(u_i - 9c \frac{u_i^2}{(1+r_i)^2} - \frac{1}{4} c \bar{g}_i^2 \right), \quad (26)$$

$$\bar{g}_{i+1} = 2^{2-d} \bar{g}_i \left(1 - \frac{3}{2} c \frac{u_i}{(1+r_i)^2} - \frac{3}{2} c \bar{v}_i - 2c \frac{\bar{g}_i}{(1+r_i)} \right), \quad (27)$$

$$\bar{v}_{i+1} = 2^{-d} \left(\bar{v}_i - 9c \bar{v}_i^2 - \frac{1}{4} c \frac{\bar{g}_i^2}{(1+r_i)^2} \right), \quad (28)$$

where the constant c is the approximate value of a

Feynman integral. The factors 2^{4-d} , 2^{2-d} , and 2^{-d} in Eqs. (26)–(28) are the same factors that we discussed before. An immediate consequence of Eqs. (25)–(28) is that, for $d > 4$, the only fixed point is the Gaussian $r^* = u^* = \bar{g}^* = \bar{v}^* = 0$, whereas for $2 < d < 4$ one finds the conventional fixed point of the uncoupled ϕ^4 theory. To order $\epsilon = 4 - d$, one then has that

$$u^* = \frac{\epsilon \ln 2}{9c}, \quad r^* = -\left(\frac{4}{9c}\right) \epsilon \ln 2; \quad \bar{g}^* = \bar{v}^* = 0 \quad (29)$$

so the coupling between the two fields is an irrelevant coupling. They decouple at the critical point, and the critical behavior of the correlation functions of the ϕ field is identical to the behavior of the correlation functions of an Ising-like ferromagnetic system.

It is important to emphasize the real nature of this decoupling of the massless field from the massive one at the antiferromagnetic critical point. Notice that we find that the origin is an infrared attractive fixed point only for the dimensional coupling constants $\bar{g} = gR^{-1}$ and $\bar{v} = vR^{-2}$, and simply from dimensional reasons one would expect them to vanish in the static limit, when $R \rightarrow \infty$. Having chosen the propagator of the massive field ψ to be 1,

what we have really done is, in the language of perturbation theory, to absorb in the coupling constants g , v factors of R^{-1} naturally provided by the massive propagator. The decoupling is then entirely due to the static limit: the massive field simply disappears from the theory in the limit $R \rightarrow \infty$. What we have achieved in this section is to show that, in order to exhibit this mechanism neatly in Wilson's formulation of the renormalization group, the recurrence relations of the RGT must be written for the dimensional parameters $\bar{g}$, $\bar{v}$. Indeed, by writing the recurrence relations for the dimensionless couplings g , v one does not find an infrared stable fixed point at $g=0$. We shall comment again on this point in Sec. IV.

IV. RENORMALIZED-FIELD-THEORY APPROACH

The purpose of this section is to use the methods of renormalized Lagrangian field theory to study the infrared stability of the theory under consideration.⁶ We must consider then two coupled scalar fields, $\phi(\vec{x})$ and $\chi(\vec{x})$, with renormalized masses m , M , respectively. At the critical temperature m tends to zero but M stays finite. Since the field χ is far from its critical point, we expect in general M to be of the order of the ultraviolet cutoff Λ . Indeed, one can easily convince oneself that M should be of the order of the inverse of the spacing between the two sublattices.

The use of renormalized perturbation theory is usually justified by emphasizing that the critical behavior of the system should be independent of the ultraviolet cutoff, so the limit $\Lambda \rightarrow \infty$ can be taken to compute quantities that are cutoff independent. Since $M \sim \Lambda$, it may look redundant at first sight to set up a formalism based on a renormalized Lagrangian field theory and keep explicitly the massive field χ . Moreover, Symanzik has given a general proof that, in a ϕ^4 theory that contains massive particles besides the massless ones, the infrared singularity structure is the same as in the theory with the massless particles only.⁸ However, we find that keeping both fields is necessary for a correct discussion of problems such as corrections to scaling and the equation-of-state antiferromagnetic system. Therefore, we shall translate in this section all the results of the previous one to the language of renormalized Lagrangian field theory. Corrections to scaling and the equation of state will be discussed at length elsewhere.

Let us start by considering the renormalized Lagrangian in $4 - \epsilon$ dimensions

$$\begin{aligned} \mathcal{L}(\vec{x}) = & -\frac{1}{2}(\vec{\nabla}\hat{\phi})^2 - \frac{1}{2}(\vec{\nabla}\hat{\chi})^2 - \frac{1}{2}m^2\hat{\phi}^2 - \frac{1}{2}M^2\hat{\chi}^2 \\ & - (1/4!)\hat{g}_1\hat{\phi}^4 - (1/4!)\hat{g}_2\hat{\chi}^4 - \frac{1}{4}\hat{g}_3\hat{\phi}^2\hat{\chi}^2 \\ & + (\text{counter terms}) . \end{aligned} \quad (30)$$

With the transformations

$$\chi = M\hat{\chi}, \quad \phi = \hat{\phi} \quad (31)$$

and

$$g_1 = \hat{g}_1, \quad g_2 = \hat{g}_2 M^{-4}, \quad g_3 = \hat{g}_3 M^{-2}, \quad (32)$$

the Lagrangian becomes

$$\begin{aligned} \mathcal{L}(x) = & -\frac{1}{2}(\vec{\nabla}\phi)^2 - (1/2M^2)(\vec{\nabla}\chi)^2 - \frac{1}{2}m^2\phi^2 - \frac{1}{2}\chi^2 \\ & - (1/4!)g_1\phi^4 - (1/4!)g_2\chi^4 - \frac{1}{4}g_3\phi^2\chi^2 \\ & + (\text{counter terms}) . \end{aligned} \quad (33)$$

Now the dimensions of the fields $\phi(\vec{x})$, $\chi(\vec{x})$ are, in mass units, $(1 - \frac{1}{2}\epsilon)$ and $(2 - \frac{1}{2}\epsilon)$, respectively. The new coupling constants have dimensions given by

$$[g_1] = \epsilon, \quad [g_2] = \epsilon - 4, \quad [g_3] = \epsilon - 2 \quad (34)$$

and the free propagators of the fields ϕ , χ are

$$(p^2 + m^2)^{-1}, \quad (1 + p^2/M^2)^{-1}, \quad (35)$$

respectively. We shall indicate by $\Gamma^{(N,N')}(p_1 \dots p_N; q_1 \dots q_{N'})$ the one-particle irreducible vertex functions corresponding to N external lines of the field ϕ with momenta p_i ($i=1, \dots, N$) and N' external lines of the field χ with momenta q_i ($i=1, \dots, N'$). The finite counter terms in the Lagrangian are fixed by the following normalization conditions:

$$\Gamma^{(2,0)}(p, -p)|_{p^2=m^2} = 0, \quad \Gamma^{(0,2)}(q, -q)|_{q^2=M^2} = 0, \quad (36)$$

$$\Gamma^{(2,0)}(p, -p)|_{p^2=\mu^2} = -(\mu^2 + m^2), \quad (37)$$

$$\Gamma^{(0,2)}(q, -q)|_{q^2=\mu^2} = -(1 + \mu^2/M^2), \quad (38)$$

$$\Gamma^{(4,0)}(p_1, \dots, p_4)|_{p_s(\mu^2)} = -g_1, \quad (38)$$

$$\Gamma^{(2,2)}(p_1 p_2; q_1 q_2)|_{p_s(\mu^2)} = -g_3, \quad (38)$$

$$\Gamma^{(0,4)}(q_1, \dots, q_4)|_{p_s(\mu^2)} = -g_2, \quad (38)$$

where μ is an arbitrary normalization mass, and $p_s(\mu^2)$ is the usual symmetric point $s = t = u = \frac{4}{3}\mu^2$ when all external legs have mass μ^2 . These normalization conditions allow us to work directly with the zero mass theory ($m=0$), which corresponds to the temperature being exactly the critical one. We shall restrict ourselves to this case from now on, and proceed to calculate the low-momentum behavior of the correlation functions at the critical point.

The vertex functions $\Gamma^{(N,N')}$ satisfy the renormalization-group equation⁹

$$\begin{aligned} & \left(\mu \frac{\partial}{\partial \mu} + \sum_{i=1}^3 \bar{\beta}_i \frac{\partial}{\partial g_i} - N\gamma_1 - N'\gamma_2 \right) \\ & \times \Gamma^{(N,N')}(p, q; \mu, M, g_1, g_2, g_3) = 0, \end{aligned} \quad (39)$$

where γ_i ($i=1, 2$) and $\bar{\beta}_j$ ($j=1, 2, 3$) are functions of the parameters of the theory and the normalization mass μ . Notice that the γ_i are dimensionless but

$\bar{\beta}_i$ have the dimension of g_i . These functions are easily determined by applying the differential operator of Eq. (39) to the vertex functions given by the normalization conditions of Eqs. (36)–(38).¹⁰

Now using the fact that $\Gamma^{(N, N')}$ is a homogeneous function of degree $4 - N - N'$, one can eliminate $\mu(\partial/\partial\mu)$ from the renormalization-group equation (39) and write it as follows:

$$\left(-\lambda \frac{\partial}{\partial \lambda} + \sum_{i=1}^3 \beta_i \frac{\partial}{\partial g_i} - M \frac{\partial}{\partial M} - N(1 - \frac{1}{2}\epsilon + \gamma_1) - N'(2 - \frac{1}{2}\epsilon + \gamma_2) + 4 - \epsilon\right) \times \Gamma^{(N, N')}(\lambda p_1, \dots, \lambda p_N; \lambda q_1, \dots, \lambda q_{N'}; g_i, \mu, M) = 0 \quad (40)$$

where

$$\beta_1 = -\epsilon g_1 + \bar{\beta}_1, \quad \beta_2 = (4 - \epsilon)g_2 + \bar{\beta}_2, \quad \beta_3 = (2 - \epsilon)g_3 + \bar{\beta}_3. \quad (41)$$

The solution of Eq. (40) is given by

$$\Gamma^{(N, N')}(\lambda p_i, \lambda q_j; g_1, g_2, g_3; M, \mu) = \lambda^{4-\epsilon-N(1-\epsilon/2)-N'(2-\epsilon/2)} \times [a_1(\lambda)]^N [a_2(\lambda)]^{N'} \Gamma^{(N, N')}(p_i, q_j; g_1(\lambda), g_2(\lambda), g_3(\lambda); (M/\lambda), \mu), \quad (42)$$

where the effective coupling constants are given by the Gell-Mann–Low equations

$$\lambda \frac{dg_i}{d\lambda} = \beta_i, \quad g_i(1) = g_i \quad (i = 1, 2, 3) \quad (43)$$

and

$$a_j(\lambda) = \exp\left[-\int_1^\lambda \frac{ds}{s} \gamma_j\left(g_i(s), \frac{M}{s}, \mu\right)\right] \quad (j = 1, 2). \quad (44)$$

We are interested in the long distance behavior

of the correlation functions, that is to say, the behavior of $\Gamma^{(N, N')}(\lambda p_i, \lambda q_j)$ when $\lambda \rightarrow 0$. The solution of the renormalization-group equations given by Eq. (42) determines the behavior in terms of the behavior of $\Gamma^{(N, N')}$ at fixed momenta but with effective coupling constants $g_i(\lambda)$ and the mass of the χ field going to infinity as M/λ . The β_i functions can be determined in perturbation theory, and with the normalization conditions given by Eqs. (36)–(38) we obtain

$$\beta_1 = -\epsilon g_1 + (3/(4\pi)^2)[\mu^{-\epsilon} g_1^2 + M^{4-\epsilon} g_3^2 B(\mu/M)] + O(g_i^3, \epsilon g_i^2), \quad (45)$$

$$\beta_2 = (4 - \epsilon)g_2 + [3/(4\pi)^2][M^{-\epsilon} g_3^2 + M^{4-\epsilon} g_2^2 B(\mu/M)] + O(g_i^3, \epsilon g_i^2), \quad (46)$$

$$\beta_3 = (2 - \epsilon)g_3 + [g_3/(4\pi)^2][\mu^{-\epsilon} g_1 + M^{2-\epsilon} g_3 C(\mu/M) + M^{4-\epsilon} g_2 B(\mu/M)] + O(g_i^3, \epsilon g_i^2), \quad (47)$$

where

$$B(x) = \frac{4}{3}x^2 + O(x^4), \quad (48)$$

$$C(x) = 2x^2 + O(x^4). \quad (49)$$

From the knowledge of the β functions one can immediately see that there is a fixed point of the renormalization group where g_2 and g_3 are zero, g_1 is of order ϵ and takes exactly the usual value associated with the one-component ferromagnetic critical point, and $M = \infty$. The behavior of the effective coupling constants as $\lambda \rightarrow 0$ is given by

$$g_1(\lambda) = \mu^{\epsilon/3} (4\pi)^2 \epsilon [1 + O(\lambda^\epsilon)] + O(\lambda^{2-\epsilon/3}), \quad (50)$$

$$g_2(\lambda) = b\lambda^{4-\epsilon}, \quad (51)$$

$$g_3(\lambda) = c\lambda^{2-(2/3)\epsilon}, \quad (52)$$

where b, c are integration constants. This behavior guarantees that the fixed point we are con-

sidering is fully infrared stable, and that the fields decouple at the critical point.

The anomalous dimensions of the fields ϕ, χ , critical exponents in statistical mechanics, are given by the values of γ_1 and γ_2 at the fixed point. As $\lambda \rightarrow 0$, they behave as

$$\gamma_1[g_i(\lambda), M/\lambda, \mu] = \epsilon^2/108 + c_1 \lambda^{2+(2/3)\epsilon}, \quad (53)$$

$$\gamma_2[g_i(\lambda), M/\lambda, \mu] = c_2 \lambda^2 + c_3 \lambda^{2+(2/3)\epsilon}, \quad (54)$$

where the c_i 's are constants. These results imply that the field χ does not have an anomalous dimension, while the field ϕ has the same anomalous dimension as in the one-component case. Therefore, the critical exponent η is the same as in the ferromagnetic case. One can also check in general that all critical exponents are the same, and that corrections to scaling are also controlled by the same subcritical coefficient w^3 of order ϵ both at the ferromagnetic and antiferromagnetic critical points.

ACKNOWLEDGMENTS

One of us (V.A.A.) wishes to thank Professor Abdus Salam, The International Atomic Energy

Agency, and UNESCO for the kind hospitality extended to him at the ICTP where most of this work was done.

¹E. Brézin, D. J. Wallace, and K. G. Wilson, Phys. Rev. Lett. 29, 591 (1972); G. M. Avdeeva and A. A. Migdal, Zh. Eksp. Teor. Fiz. Pis'ma Red. 16, 253 (1972) [JETP Lett. 16, 178 (1972)].

²E. Brézin, D. J. Wallace, and K. G. Wilson, Phys. Rev. B 7, 232 (1973).

³E. Brézin, J. C. Le Guillou, and J. Zinn-Justin, Phys. Rev. D 8, 434 (1973); 8, 2418 (1973).

⁴For a review of the applications of the renormalization group to critical phenomena, see K. G. Wilson and J. Kogut, Phys. Rep. (to be published).

⁵A good account of the two-lattice model for antiferromagnetic phase transitions is given in F. Keffer, *Encyclopedia of Physics*, (Springer-Verlag, Berlin, 1966),

Vol. XVIII/2.

⁶For a review of the renormalization group and the ϵ -expansion in the context of Lagrangian field theory, see J. Zinn-Justin, report of work prior to publication, No. DPh-T/73-79, Saclay, Oct. 1973. See also Ref. 10.

⁷E. Brézin, J. C. Le Guillou, and J. Zinn-Justin, Phys. Rev. B (to be published).

⁸K. Symanzik, (unpublished). We are grateful to Dr. J. Zinn-Justin for calling our attention to this work.

⁹M. Gell-Mann and F. E. Low, Phys. Rev. 95, 1300 (1954).

¹⁰F. Jegerlehner and B. Schroer, Acta Phys. Austriaca Suppl. 11, 389 (1973).